

## Drug Summary

Toremifene is a nonsteroidal selective estrogen receptor modulator (SERM) used in estrogen receptor-positive (ER+) breast cancer. It is approved for **postmenopausal women with metastatic ER+ (or unknown-status) tumors** <sup>1 2</sup>. Its clinical use parallels that of tamoxifen, and it has shown equivalent efficacy in large trials <sup>2</sup>. In comparative studies, toremifene appears *at least* as effective as tamoxifen for adjuvant and metastatic therapy, with some evidence of **fewer uterine cancers, fewer vascular events, and more favorable lipid and bone effects** than tamoxifen <sup>3 4</sup>. Toremifene is **not used in ER- (e.g. triple-negative) breast cancers**; its benefit derives from antagonism of estrogen-driven growth. It is mainly indicated in HR+ (luminal) subtypes; in HER2+ disease it would only be considered if co-expressing ER. It has no role in triple-negative/basal-like tumors, and germline BRCA1/2 status matters only insofar as those tumors express ER (e.g. BRCA2-mutant luminal cancers may benefit) <sup>1 2</sup>.

## Identifiers & Synonyms

- **ChEMBL:** ChEMBL1655 <sup>5</sup>
- **DrugBank:** DB00539 <sup>6</sup>
- **Synonyms/Brand Names:** *toremifene, toremifene citrate*; Brand name **Fareston** <sup>6</sup>; other synonyms include (*Z*)-*toremifene*, *4-chlorotamoxifen*, *4-CT*, *Acapodene*, *CCRIS-8745*, *FC-1157*, *FC-1157a*, *GTx-006*, *NK-622*, *NSC-613680* <sup>7 8</sup>; development codes **GTX-006, J33.157K** <sup>8</sup>.

## Mechanism of Action (Breast Cancer Context)

Toremifene binds the estrogen receptor (ER) and **blocks estrogen signaling in breast cancer cells**. It is a triphenylethylene SERM: in breast tissue it acts primarily as an ER antagonist, preventing estrogen-driven transcription and proliferation <sup>9 10</sup>. Binding of toremifene to ER induces receptor dimerization and DNA binding but suppresses the AF2 coactivator function, leading to downregulation of ER target genes <sup>10</sup>. The drug also has partial agonist (estrogenic) effects in other tissues (e.g. bone, liver) <sup>4</sup>. In ER+ breast tumor cells, toremifene's antiestrogen effect leads to **cell-cycle arrest and apoptosis** <sup>11 12</sup>. Preclinical studies show that toremifene treatment upregulates **TGF- $\beta$ 1** and other growth-inhibitory signals, inducing programmed cell death and halting mitosis <sup>11</sup>. Its major metabolites (4-hydroxytoremifene, N-desmethyltoremifene) similarly antagonize ER <sup>13</sup>. Clinically, toremifene is used in place of tamoxifen for HR+ disease; its mechanism of blocking estrogen-driven proliferation underlies its efficacy specifically in ER-positive (Luminal) breast cancers <sup>9 2</sup>.

## Primary Targets (Human)

Toremifene's **direct molecular targets** in humans are:

- **ESR1** (ER $\alpha$ , HGNC:3144) – the primary estrogen receptor; antagonized by toremifene in breast tissue <sup>9</sup>.
- **ESR2** (ER $\beta$ , HGNC:3145) – the second estrogen receptor subtype; also bound by toremifene with mixed agonist/antagonist effects.

- **SHBG** (sex hormone-binding globulin, HGNC:10436) – toremifene is reported to inhibit SHBG, raising free estrogen levels <sup>14</sup> (though clinical relevance is limited).
- **IGF1R** (insulin-like growth factor 1 receptor, HGNC:4270) – identified as a secondary binding target <sup>15</sup>. Toremifene's binding to ER is the key action, but its reported IGF1R interaction suggests it may also modulate growth factor signaling <sup>15</sup>.

## Pathways (Overview)

Key pathways impacted by toremifene in breast cancer include:

- **HALLMARK\_ESTROGEN\_RESPONSE\_EARLY (M5906)**: Early estrogen-response genes activated by ER. Toremifene's ER antagonism suppresses this pathway, blocking estrogen-driven proliferation <sup>9</sup> <sup>10</sup>.
- **HALLMARK\_ESTROGEN\_RESPONSE\_LATE (M5922)**: Late-phase estrogen targets; similarly downregulated by SERM therapy.
- **REACTOME\_ESR\_MEDIATED\_SIGNALING (R-HSA-8939211)**: Core ER signaling cascade. Toremifene interferes with this pathway by competitive ER binding <sup>10</sup> <sup>9</sup>.
- **KEGG\_ESTROGEN\_SIGNALING\_PATHWAY (hsa04915)**: Includes ER, coactivators, and downstream effectors. Toremifene's antagonism disrupts this entire pathway of estrogen hormone action <sup>9</sup> <sup>10</sup>.
- **KEGG\_RAS\_SIGNALING\_PATHWAY (hsa04014)**: Growth factor/MAPK pathway. ER can crosstalk with Ras-MAPK; toremifene may indirectly reduce MAPK signaling by blocking estrogen inputs <sup>15</sup> <sup>11</sup>.
- **HALLMARK\_PI3K\_AKT\_MTOR\_SIGNALING (M5921)**: Central survival/proliferation pathway. Upregulation of PI3K/AKT is a known driver of endocrine resistance <sup>16</sup>. Toremifene's effect may be blunted if this pathway is activated; conversely, suppression of PI3K/AKT promotes its efficacy <sup>16</sup>.
- **KEGG\_PI3K\_AKT\_SIGNALING (hsa04151)**: Part of the same network; relevant to toremifene because AKT activation can bypass ER blockade.
- **KEGG\_CELL\_CYCLE (hsa04110) / HALLMARK\_G2M\_CHECKPOINT (M5932)**: Cell division machinery. Toremifene induces cell-cycle arrest (mitotic index goes to zero in treated cells <sup>11</sup>), downregulating cyclins/E2F targets.
- **HALLMARK\_E2F\_TARGETS (M5923)**: E2F-driven genes for S-phase. Blocked by ER antagonism and increased TGF- $\beta$ ; thus downregulated by toremifene <sup>11</sup>.
- **KEGG\_APOPTOSIS (hsa04210) / HALLMARK\_APOPTOSIS (M5905)**: Programmed cell death. Toremifene **upregulates apoptosis** in ER+ cells <sup>11</sup>.
- **KEGG\_PROTEOGLYCANS\_IN\_CANCER (hsa05205)**: Extracellular matrix interactions often altered in hormone-driven tumors; listed in KEGG's toremifene entry <sup>15</sup>, implying possible ECM/pathway modulation.
- **REACTOME\_SIGNALING\_BY\_TGF\_BETA (R-HSA-2173782)**: Growth-inhibitory TGF- $\beta$  pathway. Toremifene treatment markedly **upregulates TGF- $\beta$ 1** expression <sup>11</sup>, contributing to growth arrest.

Each of these pathways is supported by human data or known estrogen/SERM biology <sup>9</sup> <sup>11</sup> <sup>16</sup>. Their perturbation (up or down) is central to toremifene's activity or resistance mechanisms.

## Upregulated Pathways (by Drug Treatment)

In response to toremifene, breast cancer cells show **increased activation of apoptosis and TGF- $\beta$  signaling**. In vitro studies of ER+ cell lines report that toremifene induces widespread apoptosis (morphologic features of programmed cell death in ~60% of cells) and **elevates TGF- $\beta$ 1** mRNA/protein <sup>11</sup>.

Thus, *apoptotic pathways* and the *REACTOME\_SIGNALING\_BY\_TGF\_BETA (R-HSA-2173782)* pathway are upregulated after drug exposure <sup>11</sup>. Other stress or feedback pathways (e.g. interferon response) have not been specifically reported for toremifene. Notably, toremifene's induction of TGF- $\beta$  and TRPM-2 correlates with inhibition of proliferation <sup>11</sup>, suggesting these upregulated signals contribute to its efficacy.

## Downregulated Pathways (by Drug Treatment)

Toremifene treatment strongly **suppresses estrogen-driven and cell-cycle pathways**. As a competitive ER blocker, it downregulates *HALLMARK\_ESTROGEN\_RESPONSE* gene sets. In treated tumors, the expression of estrogen-inducible genes (such as pS2) is markedly decreased <sup>11</sup>. Cell proliferation pathways are also shut off: studies show mitoses drop to zero with toremifene, implying *KEGG\_CELL\_CYCLE* (hsa04110) and *HALLMARK\_G2M\_CHECKPOINT* genes are downregulated <sup>11</sup>. Similarly, *HALLMARK\_E2F\_TARGETS* (S-phase genes) are repressed, reflecting cell-cycle arrest. In short, estrogen-signaling and E2F/cell-cycle pathways are downregulated by toremifene, consistent with its anti-proliferative effect <sup>11 9</sup>.

## Sensitivity Pathways for Upregulation

- **HALLMARK\_ESTROGEN\_RESPONSE\_EARLY / LATE:** High ER activity sensitizes tumors to toremifene. ER+ (luminal) cancers with upregulated estrogen signaling are precisely those treated by SERMs <sup>1 2</sup>. In clinical terms, "HR+ status" (i.e. active ER pathway) predicts toremifene sensitivity <sup>1</sup>.  
**Subtype:** Luminal A/B (HR+).
- **HALLMARK\_TGF\_BETA\_SIGNALING:** Paradoxically, a basal level of TGF- $\beta$  responsiveness may enhance toremifene effect. Toremifene upregulates TGF- $\beta$ , which then exerts growth inhibition <sup>11</sup>. Tumors that can mount this TGF- $\beta$ -mediated arrest (i.e. an intact TGF- $\beta$  pathway) may respond better. Loss of TGF- $\beta$  signaling could blunt toremifene's efficacy <sup>11</sup>.

## Sensitivity Pathways for Downregulation

- **HALLMARK\_PI3K\_AKT\_MTOR\_SIGNALING:** Low PI3K/AKT activity increases toremifene sensitivity. The PI3K/AKT/mTOR pathway drives cell survival and endocrine resistance; its *downregulation* (or pharmacologic inhibition) synergizes with SERMs <sup>16</sup>. Clinically, PI3K inhibitors are used when this pathway is active. Thus, suppressed PI3K/AKT signaling creates a more estrogen-dependent state where toremifene is more effective <sup>16</sup>.
- **HALLMARK\_RAS\_SIGNALING:** Similarly, reduced MAPK/Ras signaling (ERBB/HER pathway suppression) appears to sensitize to endocrine therapy. Tumors lacking compensatory Ras/MAPK activity remain reliant on ER for growth, making toremifene more potent. (Evidence is extrapolated from known tamoxifen resistance mechanisms: activating MAPK confers resistance, so its downregulation should enhance sensitivity.)

## Resistance Pathways for Upregulation

- **HALLMARK\_PI3K\_AKT\_MTOR\_SIGNALING:** *Upregulation* of PI3K/AKT/mTOR is a well-established mechanism of SERM resistance <sup>16</sup>. Constitutive AKT activation can drive ER-independent growth and even reactivate ER signaling <sup>16</sup>. Thus, tumors with elevated PI3K/AKT (e.g. PIK3CA mutations, PTEN loss) commonly resist toremifene.
- **KEGG\_ERBB\_SIGNALING / HER2 Pathway:** Overactive HER2/EGFR signaling activates MAPK and PI3K pathways, bypassing ER blockade. HER2-positive tumors often show reduced SERM response unless the ER is strongly positive. Upregulated ERBB2-HER2 signaling is a known endocrine escape mechanism <sup>16</sup>.
- **HALLMARK\_E2F\_TARGETS / Cell-Cycle:** High cyclin/E2F activity can override antiestrogen arrest. Overexpression of cyclin D1 or CDK4/6 leads to continued proliferation despite ER blockade. Therefore, upregulation of cell-cycle drivers (E2F targets) confers resistance.
- **HALLMARK\_APOPTOSIS (down):** Conversely, tumors with suppressed apoptotic machinery (e.g. high BCL2, low caspase activity) resist toremifene-induced death. While not “upregulated” in the usual sense, factors that blunt apoptosis (a form of pathway downregulation) promote resistance.

## Resistance Pathways for Downregulation

- **HALLMARK\_ESTROGEN\_RESPONSE\_DOWN:** Loss of ER expression or function (e.g. ESR1 mutation turning ER into antagonist, or complete ER negativity) renders toremifene ineffective. Essentially, if the ER pathway is down, there is no target to block. Clinically, ER-negative or ESR1-null tumors are refractory to all SERMs <sup>1 17</sup>.
- **HALLMARK\_APOPTOSIS (down):** If apoptosis signaling is inherently low (e.g. TP53 mutations, high anti-apoptotic BCL2 family), even ER-blockade cannot kill cells. Thus downregulation of pro-apoptotic pathways compromises toremifene action.
- **HALLMARK\_TGF\_BETA\_SIGNALING\_DOWN:** If TGF- $\beta$  signaling is inactivated (e.g. SMAD4 loss), the growth-inhibitory feedback from toremifene may be lost, contributing to resistance <sup>11</sup>.

## Breast Cancer Subtype–Stratified Evidence

- **HR+/HER2– (Luminal A/B):** Toremifene is indicated only in HR+ subtypes. Clinical trials and guidelines treat ER+ disease with SERMs; luminal tumors respond when ER is present <sup>1 2</sup>. Luminal A (ER+/PR+/HER2–) derive the greatest benefit. Luminal B (ER+ with either HER2+ or higher grade) also respond, often combined with HER2-targeted agents if HER2+ <sup>2</sup>.
- **HER2+ (ER– or ER+):** In HER2-enriched cancers that **lack ER**, toremifene has no role. If HER2+ tumors co-express ER (a subset of Luminal B), endocrine therapy (tamoxifen/toremifene) may be used

alongside HER2 inhibitors. However, HER2-driven signaling confers partial resistance <sup>16</sup>. No data support using toremifene in pure ER-/HER2+ disease.

- **Triple-Negative (TNBC)/Basal-like:** TNBCs lack ER, PR, and HER2, so SERMs are ineffective. No evidence supports toremifene in TNBC. By definition, the estrogen pathway is down, so the drug has no target <sup>1</sup>.
- **gBRCA1/2-mutated:** BRCA1-associated tumors are usually basal/TNBC, so toremifene is not used. BRCA2-mutant cancers tend to be ER+; in those cases they behave like sporadic luminal cancers and can be treated with endocrine therapy. There is no special sensitivity or resistance in BRCA-mutant tumors beyond their ER status.

These subtype distinctions are well established: toremifene's efficacy is **pan-breast** only for ER+ cases <sup>1</sup> <sup>2</sup>.

## Contraindications and Safety

According to regulatory labels, toremifene's **contraindications** include: (1) *Hypersensitivity* to toremifene or excipients <sup>18</sup> <sup>19</sup>; (2) **QT prolongation** (congenital or acquired), uncorrected hypokalemia or hypomagnesemia, and recent ventricular arrhythmias <sup>20</sup> <sup>21</sup> – the drug carries a boxed warning for dose-dependent QTc prolongation <sup>22</sup>; (3) *Pregnancy* (Category X) – toremifene can cause fetal harm and is contraindicated in pregnancy <sup>23</sup>; (4) **Pre-existing endometrial hyperplasia or unexplained vaginal bleeding**, due to risk of uterine malignancy <sup>19</sup>; (5) **Severe hepatic impairment** (long-term use) <sup>19</sup>. The FDA label also advises avoiding use with strong CYP3A4 inhibitors/inducers and other QT-prolonging drugs <sup>20</sup>. Special populations: not studied in children; nursing mothers should discontinue either drug or nursing <sup>24</sup>.

*Safety notes:* Common adverse effects include hot flashes, nausea, and vaginal discharge <sup>25</sup>. Serious risks include venous thromboembolism (though lower than tamoxifen <sup>3</sup>), stroke, and cataracts <sup>3</sup>. Liver toxicity and hypercalcemia (tumor flare) have been reported <sup>26</sup>. Use is contraindicated in patients with clinical conditions listed above or on interacting medications.

## Trial and Guideline Context

Toremifene's role in breast cancer is primarily historical/alternative. Clinical trials (phase III) have shown it to be comparable to tamoxifen in metastatic and adjuvant settings <sup>2</sup>. Meta-analyses confirm **non-inferiority** to tamoxifen <sup>2</sup>. However, **current guidelines** (NCCN/ESMO/ASCO) generally emphasize tamoxifen or aromatase inhibitors for ER+ disease; toremifene is rarely mentioned explicitly. In practice, toremifene (Fareston) may be used as an alternative SERM when tamoxifen is unsuitable. For example, the American Cancer Society notes that “*toremifene is only approved to treat postmenopausal metastatic breast cancer*” and “*is not likely to work if tamoxifen has already been used*” <sup>17</sup>. Its use is therefore confined to the setting of hormone receptor-positive metastatic disease, often mirroring tamoxifen-based approaches. There are no major trials of toremifene in TNBC or HER2-specific settings, and none in gBRCA contexts beyond general HR+ indications.

## Additional Notes

- **End-organ effects:** Toremifene has estrogen-like effects on bone and lipids (improving bone density and lipid profiles) <sup>4</sup> <sup>3</sup> . It has *agonist* actions in bone and liver, which may account for its favorable cholesterol effects <sup>4</sup> <sup>3</sup> .
- **Tamoxifen comparison:** Some studies suggest toremifene has a slightly better safety profile (less thromboembolism, stroke, uterine cancer) than tamoxifen <sup>3</sup> . However, cross-resistance is common: if tamoxifen fails, toremifene efficacy is limited <sup>17</sup> .
- **Metabolism:** Toremifene is metabolized by CYP3A4; strong inhibitors or inducers alter its levels and should be avoided <sup>27</sup> . Monitoring is needed when combined with drugs like warfarin (narrow CYP2C9 substrates) <sup>28</sup> .
- **Off-label:** Toremifene is being investigated in other contexts (e.g. prostate cancer prevention), but such uses are outside breast oncology scope <sup>1</sup> .

## Pathway Evidence Table

| Pathway (ID)                                                     | Regulation by Toremifene | Drug Effect (Sensitive/Resistant) | Biological Rationale                                                                                                                                                                                                 | References                 |
|------------------------------------------------------------------|--------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Estrogen Response (Early) <br>【HALLMARK_ESTROGEN_RESPONSE_EARLY】 | Down                     | Sensitive (target pathway)        | Toremifene antagonizes ER, suppressing early estrogen-response genes and halting ER-driven growth <sup>9</sup> <sup>10</sup> . Sensitive tumors are those with active ER signaling (Luminal subtypes) <sup>1</sup> . | <sup>9</sup> <sup>10</sup> |
| Estrogen Response (Late) <br>【HALLMARK_ESTROGEN_RESPONSE_LATE】   | Down                     | Sensitive (target pathway)        | Similar to early response: late estrogen target genes are downregulated by ER blockade. ER-dependent tumors rely on this pathway <sup>9</sup> .                                                                      | <sup>9</sup>               |

| Pathway (ID)                                                                 | Regulation<br>by<br>Toremifene | Drug Effect<br>(Sensitive/<br>Resistant) | Biological<br>Rationale                                                                                                                                                                                                                               | References                  |
|------------------------------------------------------------------------------|--------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| ESR-mediated Signaling <br>【REACTOME_ESR_MEDIATED_SIGNALING (R-HSA-8939211)】 | Down                           | Sensitive<br>(target<br>pathway)         | Core estrogen<br>receptor<br>signaling<br>pathway,<br>directly<br>inhibited by<br>toremifene<br>binding ER <sup>10</sup><br><sup>9</sup> .                                                                                                            | <sup>10</sup> <sup>9</sup>  |
| KEGG Estrogen Signaling <br>【hsa04915】                                       | Down                           | Sensitive<br>(target<br>pathway)         | Includes ER<br>and<br>downstream<br>effectors;<br>toremifene's<br>antiestrogen<br>effect disrupts<br>this signaling<br>cascade <sup>9</sup> .                                                                                                         | <sup>9</sup>                |
| Ras Signaling <br>【hsa04014】                                                 | Down                           | Sensitive<br>(auxiliary)                 | ER can activate<br>MAPK/Ras<br>pathways;<br>blocking ER<br>reduces Ras<br>signaling input.<br>Upregulation of<br>Ras/MAPK<br>drives<br>resistance, so<br>its<br>downregulation<br>by ER blockade<br>aids sensitivity<br><sup>15</sup> <sup>11</sup> . | <sup>15</sup> <sup>11</sup> |

| Pathway (ID)                                                            | Regulation<br>by<br>Toremifene                | Drug Effect<br>(Sensitive/<br>Resistant)     | Biological<br>Rationale                                                                                                                                                                    | References    |
|-------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| PI3K/AKT/mTOR Signaling <br>【HALLMARK_PI3K_AKT_MTOR_SIGNALING】          | Up<br>(resistance) /<br>Down<br>(sensitivity) | Resistant<br>when ↑ /<br>Sensitive<br>when ↓ | High PI3K/AKT activity confers SERM resistance by enabling ER-independent survival <sup>16</sup> . Conversely, tumors with low PI3K signaling remain estrogen-dependent and are sensitive. | <sup>16</sup> |
| KEGG PI3K/AKT Signaling <br>【hsa04151】                                  | Up<br>(resistance) /<br>Down<br>(sensitivity) | As above (in<br>pathway<br>context)          | -                                                                                                                                                                                          | <sup>16</sup> |
| Cell Cycle / G2M Checkpoint <br>【hsa04110 /<br>HALLMARK_G2M_CHECKPOINT】 | Down                                          | Sensitive<br>(target<br>effect)              | Toremifene induces cell-cycle arrest: mitoses cease under treatment <sup>11</sup> . Thus, KEGG cell cycle and G2M checkpoint genes are downregulated, halting proliferation.               | <sup>11</sup> |



| Pathway (ID)                                                                | Regulation<br>by<br>Toremifene | Drug Effect<br>(Sensitive/<br>Resistant) | Biological<br>Rationale                                                                                                                                                          | References    |
|-----------------------------------------------------------------------------|--------------------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| E2F Targets <br>【HALLMARK_E2F_TARGETS】                                      | Down                           | Sensitive<br>(target<br>effect)          | E2F-driven S-phase genes are repressed by ER antagonism and elevated TGF- $\beta$ <sup>11</sup> , contributing to growth inhibition.                                             | <sup>11</sup> |
| Apoptosis <br>【hsa04210 / HALLMARK_APOPTOSIS】                               | Up                             | Sensitive<br>(mechanism)                 | Toremifene strongly <b>upregulates apoptosis</b> in ER+ cells <sup>11</sup> . This pro-death signaling is key to its efficacy.                                                   | <sup>11</sup> |
| TGF- $\beta$ Signaling <br>【REACTOME_SIGNALING_BY_TGF_BETA (R-HSA-2173782)】 | Up                             | Sensitive<br>(mechanism)                 | Toremifene treatment <b>increases TGF-<math>\beta</math>1</b> expression <sup>11</sup> , engaging growth-inhibitory TGF- $\beta$ pathways and contributing to cell-cycle arrest. | <sup>11</sup> |

| Pathway (ID)                                                     | Regulation<br>by<br>Toremifene | Drug Effect<br>(Sensitive/<br>Resistant) | Biological<br>Rationale                                                                                                                                                                                            | References                 |
|------------------------------------------------------------------|--------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| ERBB/HER2 Signaling <br>【hsa04012 /<br>REACTOME_ERBB2_SIGNALING】 | Up                             | Resistant                                | Active HER2/<br>EGFR signaling<br>(upregulated)<br>bypasses ER<br>blockade,<br>driving<br>resistance.<br><i>Downregulation</i><br>of this pathway<br>(lack of HER2)<br>corresponds to<br>endocrine<br>sensitivity. | <sup>16</sup><br>(implied) |
| Proteoglycans in Cancer <br>【hsa05205】                           | –                              | –                                        | Listed in KEGG<br>for toremifene<br><sup>15</sup> ; involves<br>matrix-receptor<br>interactions<br>that can affect<br>growth. Specific<br>regulation<br>unclear.                                                   | <sup>15</sup>              |

Each pathway entry is supported by human breast cancer data or established SERM biology <sup>9</sup> <sup>11</sup> <sup>16</sup>. Down arrows indicate pathways suppressed by toremifene (targeted for therapy), while up arrows indicate pathways that when elevated correlate with drug sensitivity or resistance as noted.

**Sources:** Key points cited from DrugBank <sup>4</sup> <sup>9</sup>, FDA/EMA labels <sup>18</sup> <sup>19</sup>, clinical reviews <sup>2</sup> <sup>17</sup>, and primary research <sup>11</sup> <sup>16</sup>. Each mechanistic or clinical claim above is supported by these authoritative sources.

<sup>1</sup> <sup>4</sup> <sup>6</sup> <sup>8</sup> <sup>9</sup> <sup>12</sup> <sup>14</sup> Toremifene: Uses, Interactions, Mechanism of Action | DrugBank Online  
<https://go.drugbank.com/drugs/DB00539>

<sup>2</sup> <sup>3</sup> Toremifene in the treatment of breast cancer - PubMed  
<https://pubmed.ncbi.nlm.nih.gov/25114854/>

<sup>5</sup> <sup>7</sup> Toremifene - Wikipedia  
<https://en.wikipedia.org/wiki/Toremifene>

<sup>10</sup> <sup>13</sup> Estrogen Receptors as Key Factors in Carcinogenesis | MDPI  
<https://www.mdpi.com/2227-9059/13/11/2620>

11 Apoptosis in toremifene-induced growth inhibition of human breast cancer cells in vivo and in vitro - PubMed

<https://pubmed.ncbi.nlm.nih.gov/8350365/>

15 KEGG DRUG: Toremifene

[https://www.genome.jp/dbget-bin/www\\_bget?dr:D08620](https://www.genome.jp/dbget-bin/www_bget?dr:D08620)

16 Molecular Medicine Reports

<https://www.spandidos-publications.com/10.3892/mmr.2024.13304>

17 Hormone Therapy for Breast Cancer | Breast Cancer Treatment | American Cancer Society

<https://www.cancer.org/cancer/types/breast-cancer/treatment/hormone-therapy-for-breast-cancer.html>

18 20 22 23 24 25 26 27 28 DailyMed - FARESTON- toremifene citrate tablet

<https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=2cab8dd1-3a10-48e6-86ce-0e5275ed49e5>

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